

Parallia at #SMM4H-HeaRD 2026: ClinicalAligner26AM: A Cross-Lingual Aligner for Dataset Translation; Evidences from the MultiClinCorpus Shared Task

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Abstract

Word-level cross-lingual alignment is central to annotation projection, translation auditing, and cross-lingual faithfulness estimation, yet existing neural aligners are rarely adapted to specialized domains. In this paper, we introduce ClinicalAligner26AM, a large-context multilingual aligner model for biomedical and clinical text initialized from ClinicalEncoder26AM (Remy, 2026b). Our training recipe is inspired by AWESOME Align (Dou and Neubig, 2021). We build our soft alignment target by sharpening with Sinkhorn–Knop optimal transport a cost matrix established for parallel clinical texts and conversations through the fusion of sentence-level, phrase-level, and token-level signals. We distill this sharpened alignment matrix directly into our student aligner, by encouraging its naive cosine-based token similarity scores to match this target. At inference time, we project source-span scores through the learned token alignment matrix and decode the longest valid high-scoring span in the target text, optionally supported by MultiClinNER predictions summarized in Appendix B. We evaluate CA26AM on the MultiClinCorpus shared task, which projects Spanish clinical entity annotations into six target languages (Gallego-Donoso et al., 2026; Lopez-Garcia et al., 2026). Our two submitted systems ranked respectively first and second across all languages and entity types, with character-weighted F1 scores above 0.95 in nearly all settings.

1 Introduction

Cross-lingual word alignment has re-emerged as a practical primitive for multilingual NLP (Dou and Neubig, 2021; Chi et al., 2021). When source-language annotations are already available, alignment often offers a more direct and more reliable route to target-language supervision than training a new extractor from scratch (Jacqmin et al., 2021). This is especially true in clinical NLP, where annotated data are scarce, entity boundaries are clini-

cally consequential, and translated documents often preserve content while altering surface form (Fraile Navarro et al., 2023; Alhassan et al., 2025).

The MultiClinCorpus shared task (Gallego-Donoso et al., 2026) makes this setting explicit: given Spanish clinical texts with gold entity annotations, the goal is to transfer those annotations into post-edited English, Italian, Dutch, Swedish, Romanian, and Czech translations of these documents. The central challenge is precise localization: a target mention may expand, contract, reorder, split, or partially lexicalize relative to its source counterpart. As a result, direct span classification can be unnecessarily rigid, whereas token-level alignment naturally supports boundary variation, discontinuous realizations, nested mentions, and the transfer of span-level metadata.

In this paper, we present ClinicalAligner26AM, a cross-lingual token aligner for biomedical and clinical text derived from ClinicalEncoder26AM (Remy, 2026b). Our approach is inspired by AWESOME Align (Dou and Neubig, 2021), but adapted to the clinical domain in two key ways. First, we initialize our student and its teachers with clinically finetuned models, and focus our alignment training on parallel corpora of clinical notes and conversations. Second, instead of relying on a single bilingual embedding space, we construct a frozen alignment target from multiple complementary signals spanning sentence-level, phrase-level, and token-level semantics, as well as a positional bias.

We also introduce ClinicalAligner26AM-MCAI, a competition variant that injects supervision from manually mapped spans in the MultiClinCorpus training split (Lima López et al., 2026).

Our contribution is therefore twofold: a domain-specialized multilingual aligner for clinical text, and a simple but effective recipe for training it from frozen multi-view alignment targets.

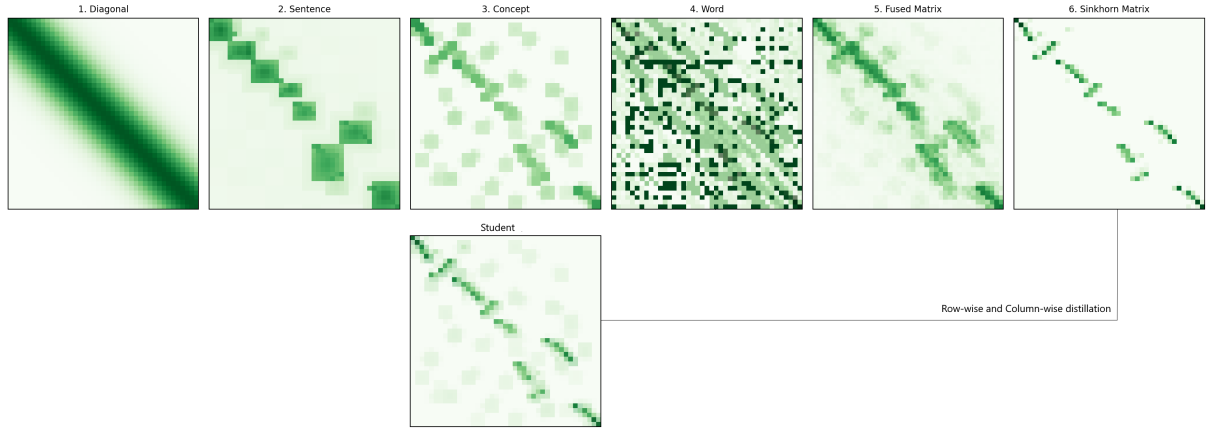


Figure 1: Training overview for ClinicalAligner26AM. We combine a diagonal positional prior with sentence-level, concept-level, and word-level similarity matrices to form a fused source–target cost matrix. Sinkhorn–Knopp optimal transport sharpens this fused matrix into a sparse soft alignment target, which is then distilled into the student aligner with row-wise and column-wise supervision.

2 Methodology

ClinicalAligner26AM is initialized from ClinicalEncoder26AM and trained as a token-level aligner on parallel clinical texts. The model is designed so that inference remains simple: source and target texts are encoded independently, their token embeddings are compared with cosine similarity, and the resulting alignment matrix is used directly for annotation projection. The difficulty lies in training this simple aligner to recover clinically meaningful correspondences without access to paired cross-attention at test time.

Our training recipe therefore separates *teacher construction* from *student learning*, as summarized in Figure 1. On the teacher side, we build a frozen source–target cost matrix that combines a weak positional prior with three complementary semantic views. The sentence-level view comes from ClinicalEncoder26AM-L2, the phrase-level view comes from ClinicalEncoder26AM-L1, and the token-level view comes from XLM-Align (Chi et al., 2021). All three teachers are frozen throughout alignment training and never receive gradients.

The positional prior reflects the structure of translated documents, which usually preserve broad discourse order even when local wording changes. We therefore assign maximum prior mass within a 50-token window around the diagonal, decay this bias up to 100 tokens, and disallow matches beyond that range. This prior is intentionally weak: it stabilizes long-document matching, but keeps the final alignment driven by semantic compatibility.

For the token-level view, we found that raw token embeddings are often too noisy for stable clinical alignment. We therefore compute token similarity with a 5-token contextualization heuristic centered on the token of interest, where the embeddings of the two preceding and the two following token are concatenated to the embedding of the main token, with strong discount factor of 0.1. This preserves token-centered behavior while reducing spurious matches caused by short tokens carrying little semantic weight.

We then sum the positional and semantic signals into an initial cost matrix and sharpen it with the Sinkhorn–Knopp algorithm (for 100 iterations) to obtain a doubly stochastic soft alignment target. This optimal-transport step converts heterogeneous similarities into a globally consistent many-to-many alignment distribution.

Our training objective consists of the distillation of this frozen target into ClinicalAligner26AM. Given independently encoded source and target sequences, the student produces a direct cosine-based alignment matrix. We train it to match the optimized teacher alignment as closely as possible, in both source-to-target and target-to-source directions. Because the matrix of cross-lingual cosine similarities is inherently not normalized, we rely on a loss combining the row-wise and column-wise cosine similarities between the naive and optimized matrices, instead of KL divergence. This objective encourages forward and backward consistency while preserving the efficiency of direct embedding comparison at test time.

Model Group	Entities	Char R	Char P	Char F1	Strict R	Strict P	Strict F1
<i>Aligners</i>							
CA26AM+MCAI	Procedure	0.96 ± 0.02	0.96 ± 0.02	0.96 ± 0.02	0.83 ± 0.03	0.82 ± 0.03	0.82 ± 0.03
CA26AM+MCAI	Disease	0.97 ± 0.02	0.97 ± 0.02	0.97 ± 0.02	0.86 ± 0.03	0.85 ± 0.03	0.85 ± 0.03
CA26AM+MCAI	Symptom	0.96 ± 0.02	0.96 ± 0.02	0.96 ± 0.02	0.82 ± 0.03	0.81 ± 0.03	0.81 ± 0.03
CA26AM	Procedure	0.95 ± 0.02	0.95 ± 0.02	0.95 ± 0.02	0.81 ± 0.03	0.80 ± 0.03	0.81 ± 0.03
CA26AM	Disease	0.96 ± 0.02	0.96 ± 0.02	0.96 ± 0.02	0.84 ± 0.03	0.83 ± 0.03	0.84 ± 0.03
CA26AM	Symptom	0.95 ± 0.02	0.95 ± 0.02	0.95 ± 0.02	0.80 ± 0.03	0.79 ± 0.03	0.80 ± 0.03
<i>Baseline without source alignment</i>							
CE26AM	Procedure	0.85 ± 0.02	0.81 ± 0.02	0.83 ± 0.02	0.74 ± 0.03	0.70 ± 0.03	0.72 ± 0.03
CE26AM	Disease	0.84 ± 0.03	0.79 ± 0.02	0.81 ± 0.02	0.72 ± 0.03	0.68 ± 0.03	0.70 ± 0.03
CE26AM	Symptom	0.79 ± 0.03	0.73 ± 0.03	0.76 ± 0.03	0.65 ± 0.04	0.60 ± 0.03	0.62 ± 0.03

Table 1: Summary of the MultiClinCorpus results of our ClinicalAligner26AM systems (Character-weighted & Strict Recall, Precision, and F1; all reported as mean ± standard deviation across the 6 languages).

4 Results

Table 1 summarizes the main outcome of the shared task. Our two aligner submissions ranked first and second overall across all target languages and all three entity types, with very small gaps between them. Character-level F1 is above 0.95 in most settings, and approaches 0.97 for several language–entity combinations, especially in English, Italian, and Romanian. These results show that token-level alignment is sufficient to recover highly accurate target spans once source annotations are provided, when an appropriate long-context in-domain neural aligner is available.

The comparison with the CE26AM baseline is particularly instructive. In our earlier MultiClinNER setting, performance was limited by the usual difficulty of mention discovery, which created visible precision issues and a larger recall–precision trade-off. Here, the source entities are already known, so the task becomes one of localization rather than detection. This substantially simplifies the problem and explains why the aligner models improve so sharply over the NER-only baseline, especially on the character-based metrics that best reflect near-miss boundary errors. For completeness, Appendix B includes a compact summary of that earlier submission, including its ranking table, optimization curve, and error profile.

The MCAI variant generally improves over the base aligner, but only modestly. This small gap is itself informative: most of the performance appears to come from the quality of the frozen multi-view alignment target rather than from task-specific span supervision alone. At the same time, the consistent edge of CA26AM+MCAI suggests that annotation-aware biasing still helps refine difficult boundary decisions.

More broadly, these results support token-level alignment as an effective mechanism for transferring exact clinical spans across translated documents. Because the method operates at the token level, it can in principle support split mentions, nested annotations, and metadata transfer even though MultiClinCorpus does not fully stress these capabilities. This same property also makes the model attractive for downstream applications such as translation auditing, error localization, and cross-lingual faithfulness assessment.

5 Conclusion

In this work, we introduced ClinicalAligner26AM, a ClinicalEncoder26AM-initialized cross-lingual token aligner for biomedical and clinical text. Our central idea is to train a simple inference-time aligner from a richer frozen teacher target built from multiple semantic views and sharpened with optimal transport. We further presented ClinicalAligner26AM-MCAI, which injects annotation-aware alignment bias from the MultiClinCorpus training split.

Across the MultiClinCorpus shared task, the two variants ranked first and second overall and achieved near-perfect multilingual entity transfer, with character-level F1 above 0.95 in nearly all settings. These results suggest that when source-language annotations are already trusted, token-level alignment is an especially effective way to turn cross-lingual transfer into a localization problem rather than a full mention-discovery problem.

More broadly, the same formulation should extend naturally to richer annotation schemas, including nested spans and metadata transfer, and to downstream settings such as translation auditing, document comparison, and summary assessment.

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A Appendix: MultiClinCorpus Full Results

Entity Type	Lang	CA26AM+MCAI		CA26AM		CE26AM	
		Char F1	Strict F1	Char F1	Strict F1	Char F1	Strict F1
PROCEDURE	cz	0.946	0.819	0.935	0.806	0.828	0.688
PROCEDURE	en	0.961	0.841	0.958	0.840	0.835	0.711
PROCEDURE	it	0.963	0.799	0.958	0.771	0.828	0.697
PROCEDURE	nl	0.916	0.770	0.907	0.755	0.811	0.692
PROCEDURE	ro	0.967	0.856	0.965	0.849	0.840	0.722
PROCEDURE	sv	0.946	0.806	0.938	0.795	0.828	0.702
DISEASE	cz	0.957	0.851	0.941	0.824	0.798	0.674
DISEASE	en	0.976	0.896	0.968	0.884	0.851	0.752
DISEASE	it	0.975	0.882	0.966	0.843	0.824	0.703
DISEASE	nl	0.932	0.820	0.924	0.804	0.793	0.678
DISEASE	ro	0.974	0.898	0.967	0.878	0.831	0.719
DISEASE	sv	0.943	0.828	0.946	0.825	0.806	0.681
SYMPTOM	cz	0.951	0.811	0.936	0.791	0.736	0.585
SYMPTOM	en	0.971	0.879	0.964	0.864	0.779	0.654
SYMPTOM	it	0.972	0.831	0.962	0.787	0.764	0.598
SYMPTOM	nl	0.921	0.768	0.915	0.763	0.726	0.564
SYMPTOM	ro	0.971	0.861	0.962	0.834	0.774	0.625
SYMPTOM	sv	0.954	0.811	0.947	0.799	0.767	0.619

Table 2: Full leaderboard summary for the Parallia submission across entity types and languages, reporting the Character-weighted and Strict F1 scores for our two aligner models (ClinicalAligner26AM-MCAI and ClinicalAligner26AM) as well as our NER-only baseline (ClinicalEncoder26AM) detailed in Annex B.

B Appendix: MultiClinNER Support Submission

This appendix summarizes the MultiClinNER submission whose predictions are optionally used during CA26AM decoding. These predictions performed particularly well in the leaderboard, especially in recall.

Entity Type	Lang	Char R rk	Char P rk	Char F1 rk	Strict R rk	Strict P rk	Strict F1 rk	Char F1	Strict F1
PROCEDURE	cz							0.8276	0.6880
PROCEDURE	en							0.8348	0.7110
PROCEDURE	es							0.8476	0.7536
PROCEDURE	it							0.8275	0.6969
PROCEDURE	nl							0.8112	0.6922
PROCEDURE	ro							0.8396	0.7222
PROCEDURE	sv							0.8275	0.7015
DISEASE	cz							0.7983	0.6737
DISEASE	en							0.8510	0.7520
DISEASE	es							0.8431	0.7512
DISEASE	it							0.8235	0.7027
DISEASE	nl							0.7926	0.6778
DISEASE	ro							0.8307	0.7189
DISEASE	sv							0.8058	0.6809
SYMPTOM	cz							0.7360	0.5847
SYMPTOM	en							0.7791	0.6542
SYMPTOM	es							0.7914	0.6696
SYMPTOM	it							0.7637	0.5976
SYMPTOM	nl							0.7258	0.5637
SYMPTOM	ro							0.7743	0.6248
SYMPTOM	sv							0.7666	0.6187

Table 3: Full leaderboard summary for the MultiClinNER submission based on ClinicalEncoder26AM. The medal-style rank display is intentionally preserved from the original system paper.

In the next page, we summarize the pre-training, NER finetuning of our ClinicalEncoder26AM model. We also provide a detailed analysis of our results in Table 3, and refer to Remy (2026b) for more details.

Pretraining. ClinicalEncoder26AM is a clinically post-trained multilingual encoder built on top of BGE-M3 (Chen et al., 2024) and aligned with a clinically relevant latent space at multiple levels, including sentence-, context-, and concept-level semantics, following the Diagnosable ColBERT methodology (Remy, 2026a). Its backbone follows an XLM-RoBERTa-style architecture (Conneau et al., 2020) with 24 Transformer layers, a hidden size of 1024, 16 attention heads, and support for up to 8192 tokens in a single pass. Its post-training combines synthetic clinical notes, patient-doctor conversations, curated translations, and annotated biomedical supervision such as MedMentions (Mohan and Li, 2019). During this stage, supervision is applied at multiple granularities, including a ColBERT-style late-interaction objective. This multi-view adaptation is designed to preserve the multilingual alignment inherited from BGE-M3 while making token embeddings more clinically meaningful and diagnosable. Because the encoder can project tokens into ontology-linked concepts such as UMLS (Bodenreider, 2004) and SNOMED CT (Donnelly, 2006), it offers broad semantic coverage that is useful both for mention linking and for identifying clinically plausible evidence spans.

Methodology. For the MultiClinNER shared task, ClinicalEncoder26AM is finetuned as a lightweight BIO tagger for procedure, disease, and symptom extraction. Indeed, while good semantic linking and good boundary detection are related, they are not the same problem. A clinically grounded encoder may already place tokens near the right ontology concepts or clinically similar contexts, yet exact span extraction still requires deciding where a mention begins and ends in free text. For that reason, the original system uses a simple supervised tagging head on top of the pretrained encoder rather than relying on ontology projection alone. More specifically, the encoder outputs are passed through a two-layer CNN block with window size 5 and hidden size 1024 before the final token-classification layer. This CNN design is intended to capture the short-range contextual cues that are especially important for span boundary detection, while keeping the overall tagging architecture lightweight. In practice, most documents fit in a single long-context forward pass, and this CNN-augmented tagging head learns local BIO transitions on top of semantically rich token representations.

Results. Our MultiClinNER submission scored exceptionally well, finishing in the Top 5 across nearly all languages, entity types, and metrics; but the system was clearly recall-oriented. It ranked first on character-weighted recall in most languages across all entity types, which indicates that the clinically post-trained encoder localized relevant evidence broadly and transferred well across languages, including Czech. Its weaker side was precision, especially on strict metrics, where long or clinically compositional mentions were sometimes broken into multiple nearly adjacent spans. Our ablation study demonstrated the usefulness of the pre-training, as the clinically post-trained model converged faster and stayed at lower loss than the base BGE-M3 model during downstream finetuning, as shown in Figure 3. This supports the view that the post-training step does not merely improve retrieval-style semantic matching, but also provides a better initialization for downstream span-level and token-level information extraction.

Reusability. In the present paper, we reuse these predictions only as auxiliary evidence during CA26AM decoding. Concretely, they help bridge difficult cases where the alignment scores identify the right region but leave small gaps between salient target tokens. The orthogonality of the two approaches made the system useful as an auxiliary source of "inside-a-span" signal for CA26AM decoding, but not strong enough to replace alignment as the primary transfer mechanism.

Conclusion. In the MultiClinNER shared task, ClinicalEncoder26AM demonstrated that clinically grounded multilingual token representations can transfer effectively to clinical named entity recognition with only a lightweight supervised head. Its strongest pattern was broad multilingual recall: the model consistently localized clinically relevant evidence across languages, even when precision and exact boundary control remained less strong. This profile suggests a pragmatic division of labor. The clinically post-trained encoder provides strong semantic coverage, while explicit span boundaries are learned by a lightweight downstream head. That is also why these predictions remain useful in the present pipeline: we do not use them as the primary transfer mechanism, but as complementary evidence in cases where alignment identifies the right target region yet leaves locally fragmented support that would benefit from merging.

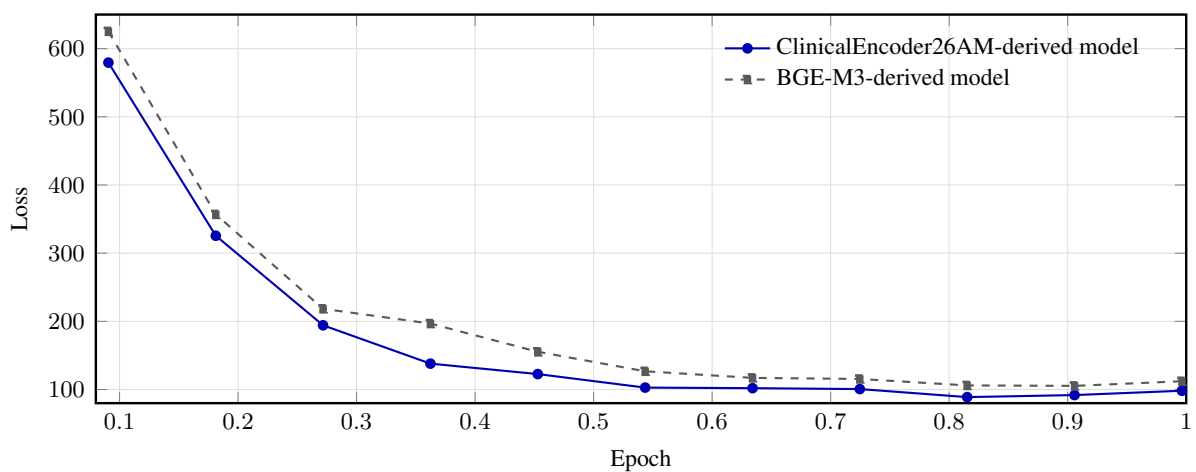


Figure 3: Training loss over the first epoch for the ClinicalEncoder26AM-derived and BGE-M3-derived MultiClinNER taggers. The clinical post-training initialization converges faster and remains at lower loss through most of finetuning.